



# Pool boiling heat transfer of saturated water on rough surfaces with the effect of roughening techniques

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## ABSTRACT

The effect of surface roughness on pool boiling heat transfer was experimentally investigated in the saturated water. Eight testing surfaces were fabricated on plain copper through three preparation methods including random polishing, unidirectional polishing, and femtosecond laser machining. The surface roughness in the present study is characterized by the areal (three dimensional, 3D) roughness parameter. In the two dimensional (2D) profile roughness analysis, we know that a profile can be separated into the large-scale component (waviness) and the short-scale component (roughness). The arithmetical mean height of the profile, the waviness and the roughness are denoted as  $Pa$ ,  $Wa$ , and  $Ra$ , respectively. However, there are no such kind of parameters defined in the 3D roughness analysis. To distinguish them, we define  $Sa_p$  and  $Sa$  in this study, corresponding to  $Pa$  and  $Ra$  in the 2D roughness parameters. The pool boiling experiments show that a rougher surface has higher heat transfer coefficient than a smoother one only within the same surface preparation method, if the surface roughness is characterized by  $Sa_p$ . For the surfaces prepared by the different techniques, the laser processed rough surface with  $Sa_p = 3.40 \mu\text{m}$  could have deteriorated heat transfer coefficient compared with the relatively smooth one polished by the 180 grit sandpaper with  $Sa_p = 1.29 \mu\text{m}$ . However, the different trend is observed when the surface roughness is characterized by  $Sa$ . By applying a standardized filtration process, the large-scale component of the surface can be removed, and the areal arithmetical mean height calculated from the remaining part is denoted as  $Sa$ . The boiling curves are found to shift monotonically to the left as  $Sa$  becomes larger, regardless of the surface preparation methods. The enhancement in heat transfer coefficient of rougher surfaces is attributed to a combined result of larger nucleation sites, smaller departure bubbles and higher bubble departure frequencies, according to the visualization study. On the other hand, the critical heat flux (CHF) does not show a clear trend with the surface roughness when the different surface preparation methods are involved, no matter  $Sa_p$  or  $Sa$  is used. The measurements of static contact angle infer that the CHF is not only dependent on surface roughness, but also highly related to surface wettability. A correlation is developed to well predict the heat transfer coefficient from the experiments as a function of  $Sa$ , especially in the fully developed nucleate boiling region.

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## 1. Introduction

Pool boiling, as a passive two-phase cooling method, is expected to be a solution for the thermal management of devices with high heat dissipation rate. The basic heat removal mechanism of pool boiling is so-called bubble nucleation, which is found to highly depend on the surface-coolant interaction. The effects of surface characteristics on pool boiling have attracted a lot of atten-

tion from researchers. Among these surface characteristics, surface roughness and contact angle are the two factors influencing the pool boiling heat transfer most. Although the surface roughness has been extensively investigated for several decades, its effect on pool boiling heat transfer is still not fully understood. The roughened surface is usually believed to enhance boiling heat transfer by providing more nucleation sites and/or larger effective heat transfer area [1–4]. Therefore, the surface roughness is normally characterized by arithmetic mean roughness  $Ra$  or effective heat transfer area ratio  $r$  (the ratio of the actual heat transfer area and its projected area) in the previous studies. Chu et al. [5] systemat-

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### List of Symbols

|                  |                                                 |
|------------------|-------------------------------------------------|
| $C, C_1, C_2$    | constant                                        |
| $C_p$            | specific heat                                   |
| $C_{sf}$         | Rohsenow constant                               |
| $g$              | gravity                                         |
| $h$              | heat transfer coefficient                       |
| $Ja$             | Jakob number                                    |
| $k$              | thermal conductivity                            |
| $L$              | latent heat                                     |
| $Nu$             | Nusselt number                                  |
| $Pr$             | Prandtl number                                  |
| $q$              | heat flux                                       |
| $r$              | effective heat transfer area ratio              |
| $Ra$             | arithmetical mean height of roughness component |
| $Rq$             | root-mean-square of roughness component         |
| $Sa$             | extension of $Ra$ to a surface                  |
| $Sa_p$           | extension of $Pa$ to a surface                  |
| $T$              | temperature                                     |
| $t$              | time                                            |
| $\Delta T_{sat}$ | wall superheat                                  |
| $\Delta x$       | distance                                        |
| $\mu$            | dynamic viscosity                               |
| $\sigma$         | surface tension                                 |
| $\rho$           | density                                         |
| $\theta$         | contact angle                                   |

### Subscripts

|       |            |
|-------|------------|
| $l$   | liquid     |
| $sat$ | saturation |
| $v$   | vapor      |
| $w$   | wall       |

### Superscripts

|        |                       |
|--------|-----------------------|
| $n, s$ | correlation exponents |
|--------|-----------------------|

ically investigated the pool boiling heat transfer performance on surfaces with the effective heat transfer area ratio  $r$  ranging from 1.79 to 5.94. Based on their experimental results, the CHF increased monotonously with  $r$  due to roughness-amplified capillary pumping. On the other hand, the heat transfer coefficient did not show a clear trend with  $r$ . Moita et al. [6] also found that surfaces with larger  $r$  could not assure a higher boiling heat transfer coefficient. They demonstrated that larger  $r$  could indeed increase the liquid-solid contact area, hence improving the heat transfer coefficient. However, sometimes it also caused the early formation of vapor blanket due to the severe interaction mechanism and coalescence effect of the nucleation bubbles. This phenomenon would deteriorate the heat transfer coefficient and even overcome the benefits of the increased liquid-solid contact area. Compared to the effective heat transfer area ratio  $r$ , the arithmetic mean roughness  $Ra$  is used more frequently to characterize the surface roughness. Many researchers visualized increased number of nucleation sites on surfaces with larger  $Ra$ , which is believed responsible for the enhancement of heat transfer coefficient [3,7,8]. However, some different observations are also reported. Benjamin and Balakrishnan [9] found the number of nucleation sites was not consistently increased with  $Ra$ . It decreased at first and then increased, as the  $Ra$  became larger. Kozitskii [10] indicated that the nucleation boiling heat transfer coefficient increased with larger  $Ra$  up to a point ( $Ra = 0.95 \mu\text{m}$ ), beyond which no improvement occurred. Similar trend was also reported by Jabardo [11] and El-Genk and Suszko [12].

The major obstacle to fully understand the effects of surface roughness on pool boiling comes from the large variances in the

reported data. Generally, these inconsistent experimental data are considered as a result of various surface roughening techniques, roughness measurement uncertainties and different experimental conditions such as working fluid, surface material and surface wettability. Jones et al. [7] conducted pool boiling experiments to investigate the effect of surface roughness on an aluminum surface in water and FC-77 respectively. They observed totally different effects of the surface roughness on the heat transfer coefficient in the two working fluids, even though the other experimental conditions were controlled the same. Kim and his group studied the effect of surface roughness on pool boiling heat transfer on a superhydrophilic [8], a moderate-wetted [4] and a hydrophobic surface [13], respectively. On the superhydrophilic and moderate-wetted surfaces, the nucleate boiling heat transfer coefficient was enhanced as the surface roughness increased, even though several cases exhibited deviation from the overall trends. However, the experiments on the hydrophobic surface showed totally different results. On the hydrophobic surface, the nucleate boiling heat transfer coefficient of a rougher surface was initially higher than that of a smoother one but decreased rapidly to a very low value in moderate and high heat flux regime, leading to an early occurrence of CHF. They explained that such trend was a result of the early formation of vapor blanket on the hydrophobic surface, same as that demonstrated by Moita et al. [6].

The surface roughening techniques are essential in studying the roughness effects on pool boiling. Researchers have emphasized that the surface roughness is meaningless without specifying the surface roughening technique. Berenson [14] found the lapped surface had higher heat transfer coefficient than the emery-polished one, although the lapped surface was smoother. Vachon et al. [15] prepared surfaces by unidirectionally polishing and chemically etching. The heat transfer coefficients of surfaces with these two surface roughening techniques were different even though the  $Rq$  (root-mean-square roughness) was similar. Other researchers also pointed out the necessity of using a consistent surface roughening technique when studying the effect of surface roughness on pool boiling [16,17]. The reason is that surfaces prepared by some roughening techniques have higher tendency to bubble nucleation, regardless of the surface roughness. Besides, the standardized surface roughness parameters, such as  $Ra$  and  $Rq$ , are statistical average values which cannot indicate the actual surface texture, especially the number of nucleation cavities. A smooth surface indicated by small  $Ra$  or  $Rq$  can also have a lot of cavities for nucleation. Many researchers reported the inadequacy of using an individual statistical parameter to characterize the surface roughness for pool boiling [2,15,17,18]. In fact, Berenson believed that a "rougher" surface in pool boiling should be that has more cavities necessary for nucleation, but not the one with a larger value of  $Ra$  or  $Rq$ . Some researchers developed methods to measure the surface active cavity size, in order to determine the potential nucleation sites quantitatively [19–22]. However, their methods are based on the simplistic treatment of the surface texture and wetting phenomenon. The accurate quantitative prediction of the nucleation sites is still unrealizable. Researchers still need to characterize the surface roughness with the aid of standardized roughness parameters.

The present study aims to investigate the effect of surface roughness on copper surfaces. The experiments were conducted in the saturated water at atmospheric pressure. A total of eight testing surfaces were prepared by three different surface roughening techniques, respectively. The different surface roughening techniques can introduce different features of surface texture. These features can be effectively eliminated by applying a novel pre-processing method before surface roughness measurement. The trends in the nucleate boiling heat transfer coefficient and CHF were discussed and the bubble dynamics was studied using a high

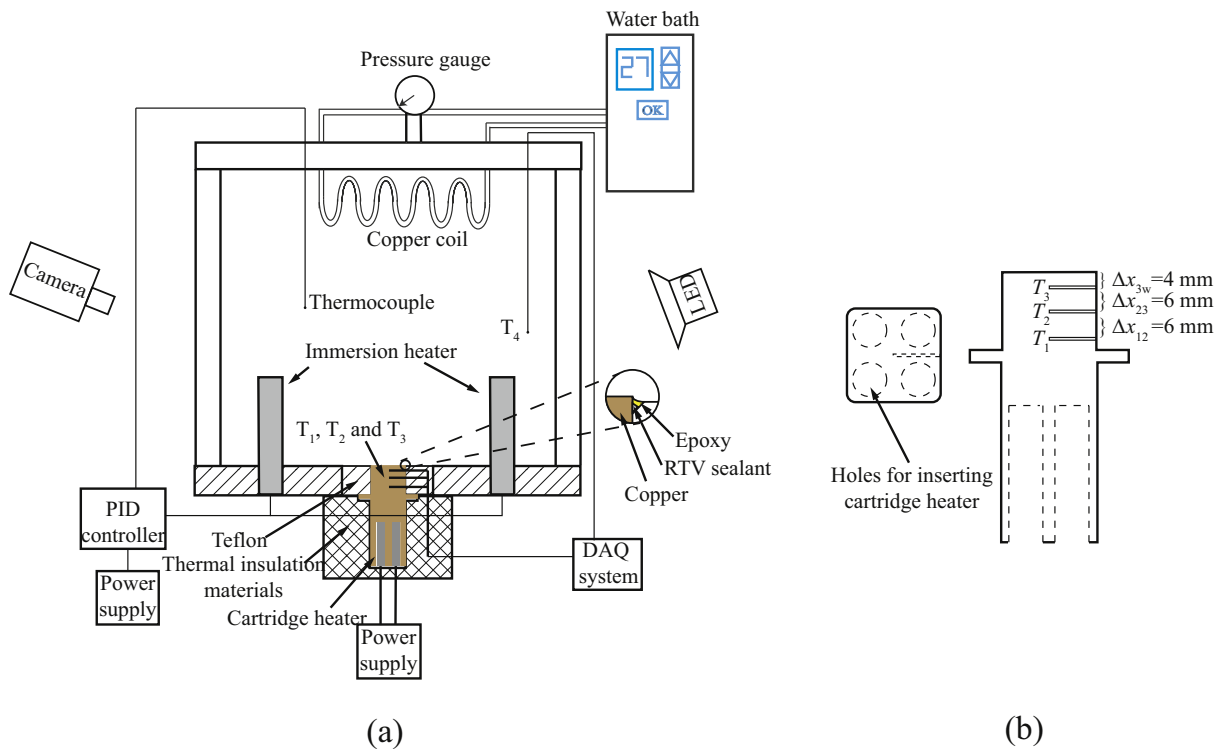


Fig. 1. (a) Schematic diagram of the experimental apparatus, (b) detailed drawing of the heating block.

speed camera in order to help understand the boiling heat transfer performance. Based on the experimental data, a correlation including the effect of surface roughness was proposed by using the arial roughness parameter  $S_a$ .

## 2. Experimental setup and procedures

### 2.1. Experimental setup

The schematic diagram of the experimental setup is shown in Fig. 1(a). The whole apparatus consists of a boiling chamber, a testing part, a temperature measurement system and a visualization system. The boiling chamber has a transparent side wall and top cover made of Acrylic in order to perform visualization study. The base plate of the boiling chamber is made of Teflon for the thermal insulation purpose. Experiments were conducted in this chamber using distilled water under atmospheric pressure. A pressure gauge is connected to the chamber in order to monitor the internal pressure. A copper coil, together with water bath, is performed as a vapor condenser. The condenser is able to condense the vapor generated in the chamber and maintain the liquid level constant. Four immersion heaters are installed inside the boiling chamber to provide heat to the liquid. These immersion heaters are connected to a proportional-integral-derivative (PID) controller, which can control the liquid temperature by detecting it with a K-type thermocouple. In this experiment, the liquid temperature is kept at around  $99.5^\circ\text{C}$ , which is the saturation temperature at atmospheric pressure. In the testing part, a copper block is surrounded by the thermal insulation material. The testing surface on the top of the copper block has a dimension of  $20 \times 20 \text{ mm}^2$ . The copper block is heated by four heaters with the capability of providing a maximum heat flux of  $150 \text{ W/cm}^2$ . Three thermocouples with hot junction diameter of  $0.5 \text{ mm}$  are aligned  $4 \text{ mm}$ ,  $10 \text{ mm}$  and  $16 \text{ mm}$  from the testing surface in the center of the copper block with holes of  $0.6 \text{ mm}$ , as shown in Fig. 1(b). During the experiments,

the bubbles were prior to generate at the edge of the testing surface. Methods of eliminating the edge nucleation sites had been performed. The one shown in Fig. 1 is proved to efficiently reduce the edge nucleation sites. The space between the copper and the teflon holder is first filled with RTV sealant and then covered by a fluidic thermal insulation material 3M Epoxy DP 460. There is still nucleation sites at the edge using this method due to the different thermal expansion rate of the copper, RTV sealant and the epoxy. But this method could effectively reduce the number of the edge nucleation sites to an acceptable range. The temperature measurement system includes four K-type thermocouples, a data acquisition system (Agilent 34972A LXI Data acquisition Unit) and a computer. All the thermocouples were calibrated using a dry block calibrator (Techne Ucal 400+) with an accuracy of  $0.2^\circ\text{C}$ . The visualization system includes a LED light source (Leica CLS150 LED), a high speed camera (Fastec Imaging HiSpec 2G Mono) and a computer. Bubbles at the boiling surface were captured and recorded by the visualization system at a frame rate of  $2000\text{fps}$  during the experiments.

### 2.2. Surface preparation

The modified surfaces in this experiment were prepared by two different methods. Before applying the surface preparation methods, the surface was first randomly polished with 2400 grit sandpaper (Struers Silicon Carbide Grinding Paper) and cleaned by sonicating in distilled water. The treated surface served as base surface in the experiment. The base surface was then roughened by unidirectional polishing with rougher sandpapers (80, 120, 180, 600, and 1200 grit) or femtosecond laser processing. A total of eight samples were prepared, including five unidirectionally polished surfaces, two laser processed surfaces and a base surface as reference. Two important surface characteristics, surface roughness and static contact angle, were measured on the testing surfaces. The surface roughness was measured by a 3D optical surface metrology system, Leica DCM-8. The measurement results are

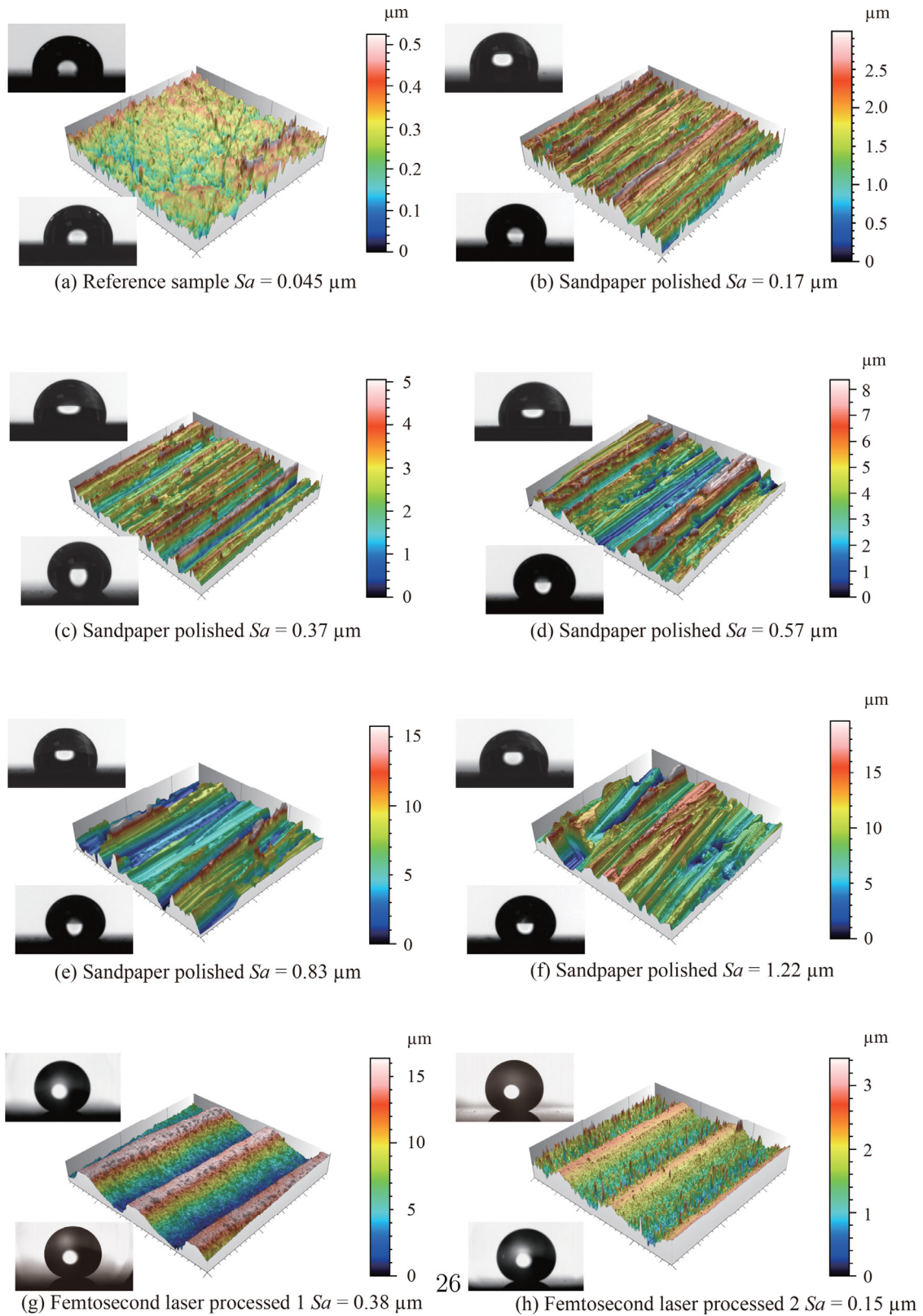


Fig. 2. Measurement results of the surface topography and surface wettability.

illustrated in Fig. 2. It can be found that the unidirectionally polished surface has a lot of irregular scratches, while the femtosecond laser processed samples have periodically fluctuating surface texture as shown in Fig. 2(g) and (h). Based on the specification standards from ISO (international Organization of Standardization),

the raw measured data is defined as surface topography which contains all the surface features, while what we concerned is only part of them, i.e., surface roughness. Therefore, the following pre-processing is applied before analyzing the surface roughness parameters.

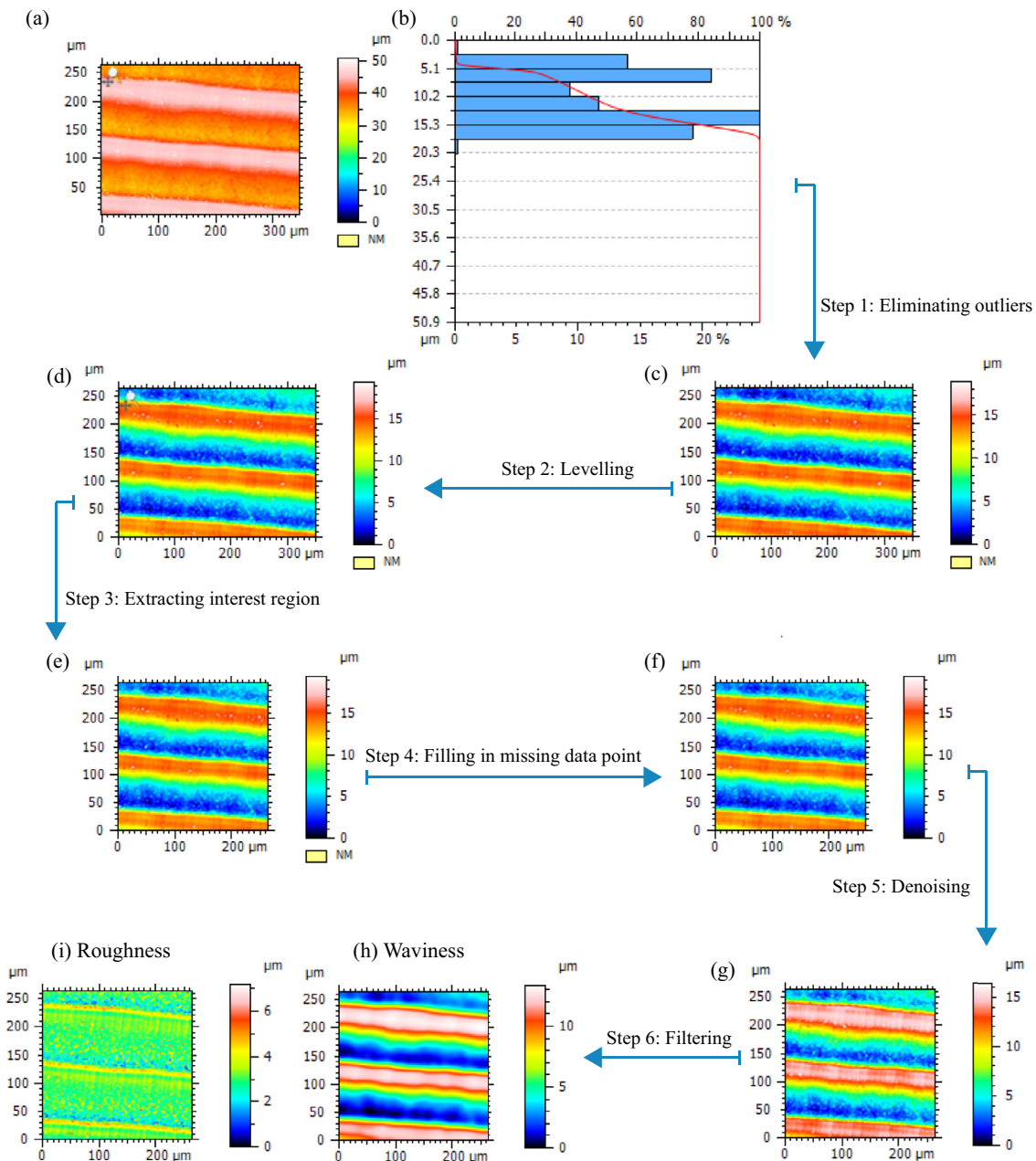
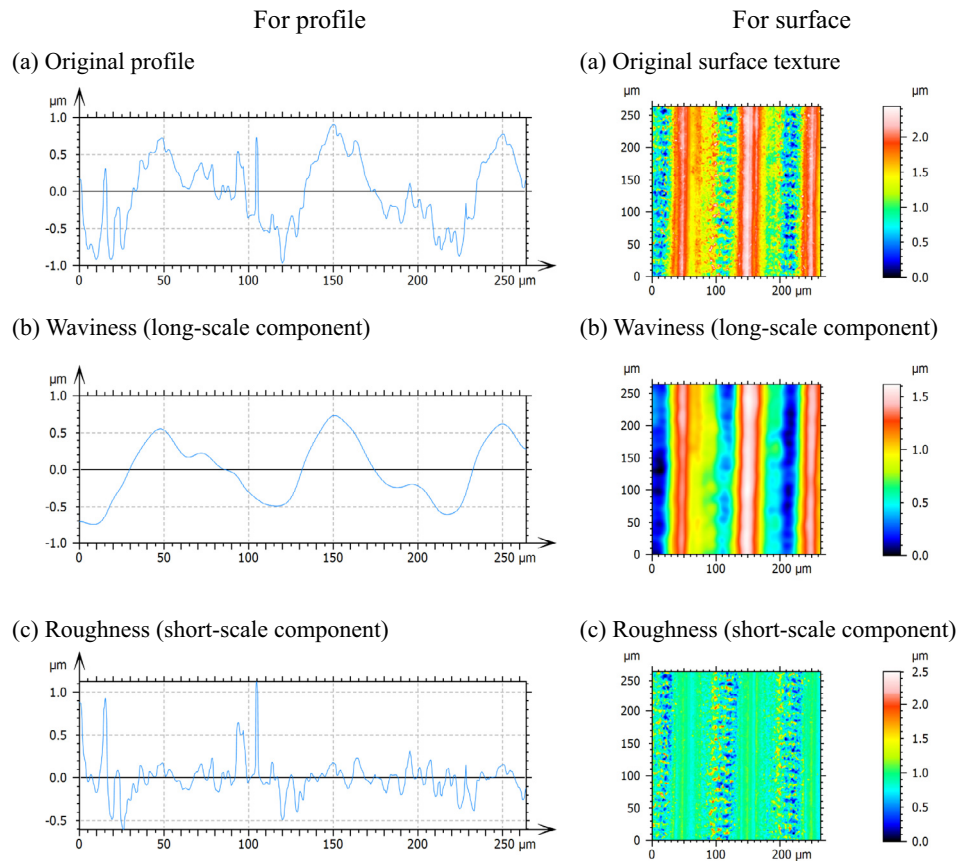


Fig. 3. Example of the pre-processing before the roughness parameter evaluation.

As shown in Fig. 3, Fig. 3(a) is the raw measured surface. It can be found that some bad points compress the surface dynamic, impeding us from obtaining detained information on the surface. This can be seen more clearly on the depth distribution of the measured surface (Fig. 3(b)). A few number of deep holes covers most of the depth range (from 20.3  $\mu\text{m}$  to 50.9  $\mu\text{m}$ ), while the main body of the surface (over 99.9% number of point) locates only at 0–20.3  $\mu\text{m}$ . This phenomenon is an inherent drawback of the non-contact instrument, as explained by the manufacturer. A spatial filter operator, with a median denoising filter  $3 \times 3$ , could eliminate the bad points (Step 1). The surface is then leveled by subtracting the least square plane from the whole surface area (Step 2). The leveled surface (Fig. 3(d)) is a rectangular one. A square area is ex-

tracted from the leveled surface for better analysis (Step 3). During the measurement, some data points were not measured due to various reasons. These missing points are filled in using a smooth shape calculated from the neighboring points (Step 4). After that, a median smoothing filter  $7 \times 7$  is applied to the surface in order to eliminate noise and improve image quality (Step 5). For surface texture, there are two geometrical product specifications published by the International Organization for Standardization (ISO), i.e., ISO 4287 and ISO 25178. The ISO 4287 specifies the terms, definitions and surface texture parameters based on profile method (2D), while the ISO 25178 collects the international standards relating to surface texture based on areal method (3D). The denoised surface after Step 5 can be treated as the S-F surface in ISO 25178,



**Fig. 4.** Illustration of applying the robust gaussian filter (25  $\mu\text{m}$  cut-off) on the profile and surface, respectively.

which is a surface obtained after applying an F-operator (a filter to correct tilt and suppress the nominal surface texture characterization). The areal primary parameters  $Sa_p$  (corresponds to  $Pa$  in ISO 4287) and the effective heat transfer area ratio  $r$  can be calculated from this surface. As shown in Fig. 3(g), the S-F surface still consists of long wavelength component (waviness) and short wavelength component (roughness). For pool boiling, the waviness part is obviously too large for nucleation cavities which are only in micro-scale. Therefore, the denoised surface is split into waviness components (Fig. 3(h)) and roughness components (Fig. 3(i)), by applying an areal robust gaussian filter with a cut-off of 25  $\mu\text{m}$ . The surface roughness parameters are then calculated from the roughness components (Fig. 3(i)) according to ISO 25178 standard.

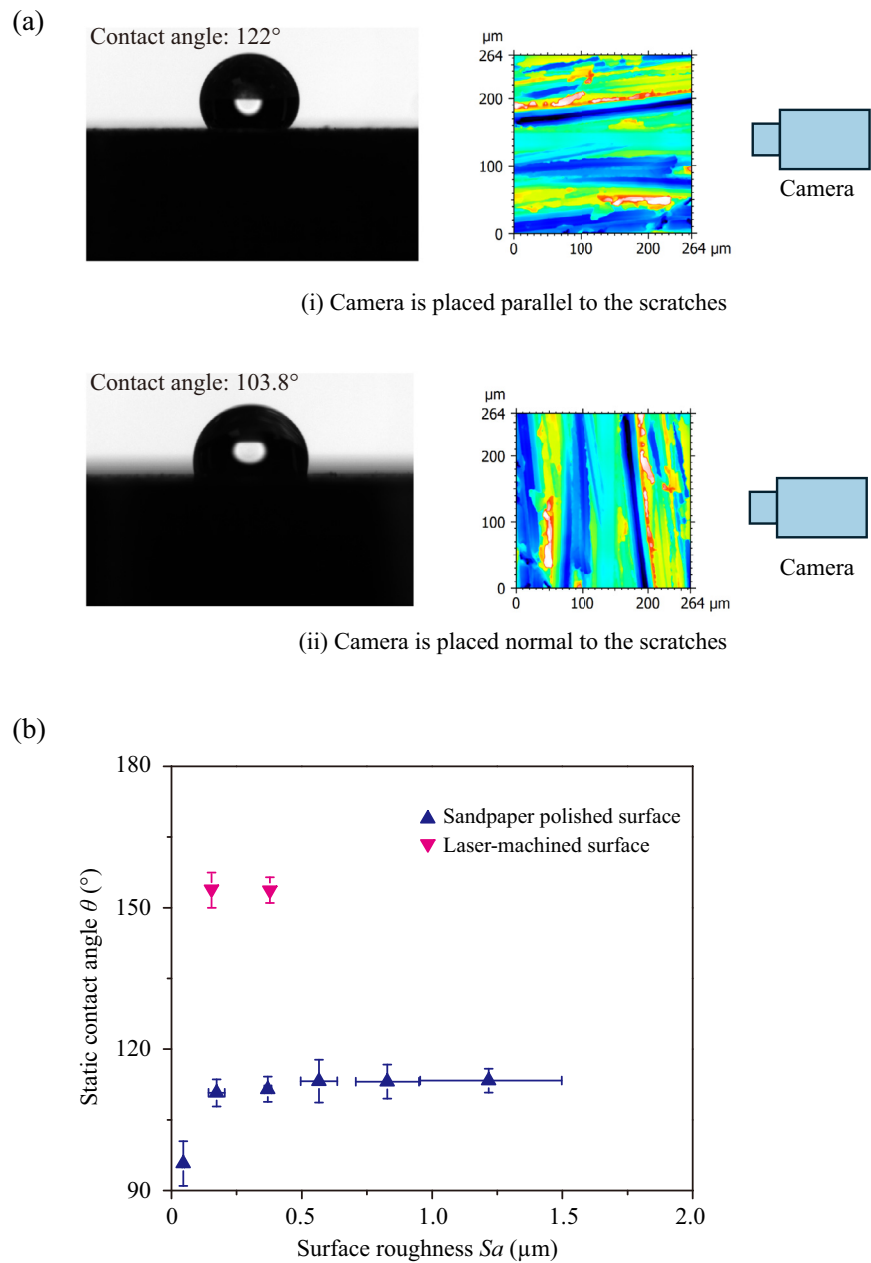
In the filtration process, selecting appropriate cut-off value (the limit wavelength between the long-scale component and short-scale component) is critical in order to separate roughness from waviness properly. There is not a generally accepted criterion for determining the cut-off value, especially for the areal filter. The choice of cut-off value is very empirical and highly dependent on the practical application. In present study, the cut-off is determined as a result of many attempts, and a cut-off value of 25  $\mu\text{m}$  is finally selected in order to assure that the irrelevant long-scale component is well removed and the short-scale component which has similar magnitude with the nucleation cavities can be remained as much as possible, as shown in Fig. 4. In the ISO 25178 standard, both the areal arithmetical mean height calculated from Fig. 3(g) and that calculated from Fig. 3(i) are denoted as  $Sa$ . To distinguish them, the former one is denoted as  $Sa_p$  in this study. All the measured results are summarized in Table 1. It should be noticed that all the parameters are calculated by averaging the results of three scans in different locations on each surface.

**Table 1**

Summary of the roughness parameter measurements on the testing surfaces.

| Sample                        | $r$  | $Sa_p$ ( $\mu\text{m}$ ) | $Sa$ ( $\mu\text{m}$ ) |
|-------------------------------|------|--------------------------|------------------------|
| Reference sample              | 1    | 0.06                     | 0.045                  |
| Sandpaper polished 1200 grit  | 1.02 | 0.25                     | 0.17                   |
| Sandpaper polished 600 grit   | 1.06 | 0.53                     | 0.37                   |
| Sandpaper polished 180 grit   | 1.08 | 1.29                     | 0.57                   |
| Sandpaper polished 120 grit   | 1.13 | 1.87                     | 0.83                   |
| Sandpaper polished 80 grit    | 1.24 | 3.48                     | 1.22                   |
| Femtosecond laser processed 1 | 1.15 | 3.40                     | 0.38                   |
| Femtosecond laser processed 2 | 1.02 | 0.42                     | 0.15                   |

The static contact angle is measured by an optical contact angle measuring and contour analysis system, DataPhysics OCA 15EC. The volume of the sessile droplet in the measurement is 10  $\mu\text{L}$ . It should be noticed that the static contact angle of the testing surfaces (except the reference sample) is different when the camera is aligned in different directions. Therefore, the measurement was performed in two different directions, i.e., camera was placed either parallel or normal to the scratches, as shown in Fig. 5(a). The measurement results of two directions on each surface are illustrated in Fig. 2. The static contact angle is an average value of the measurements in two directions. And for each direction, the measurement was repeated three times at different locations. In general, the static contact angle of a testing surface was calculated by averaging six measurement results which were measured from three different locations and two directions. The results of the contact angle measurement as a function of surface roughness  $Sa$  are shown in Fig. 5(b). All the testing surface are hydrophobic. However, their hydrophobicity shows significant disparity due to dif-



**Fig. 5.** (a) Static contact angle measurement at different directions, (b) relation between static contact angle and surface roughness  $Sa$ .

ferent surface preparation methods. The static contact angle of the laser processed surfaces is much larger than that of the sandpaper polished surfaces. For the polished surfaces, the static contact angle initially increases with surface roughness and remains a constant for further increase in surface roughness. This relationship between the static contact angle and the surface roughness is different from the results reported in previous literature [13,23,24]. The specific surface topography in the experiment can be a possible reason of this trend. As we know, the static contact angle highly depends on the spreading effect of the sessile drop on the testing surfaces. Rougher surface with larger scratches could obstruct droplet from spreading on the surface, especially in the direction normal to the scratches. Therefore, the static contact angle is increased as the surface becomes rougher. This increase does not occur anymore when the surface roughness  $Sa$  is larger than 0.57

$\mu\text{m}$ . Beyond which, the obstruction effect of the droplet spreading has encountered a threshold value. Further increase in surface roughness will not cause the surface more hydrophobic. On the other hand, for the femtosecond laser processed surfaces, the static contact angles of both surfaces were around  $153^\circ$ . The surface roughness had negligible effect on the static contact angle.

### 2.3. Experimental procedures

The boiling chamber was filled with distilled water to a level about 150 mm above the testing surface. Before the experiment, the water was first heated up by the immersion heaters and boiled for around one hour in order to release the non-condensable gases. After degassing, the PID controller was set to  $99.5^\circ\text{C}$  and the liquid was maintained at saturation state. Once the internal pressure

and liquid temperature got stabilized, a settled voltage was applied to the test heaters. The wall temperature would experience an initial increase and gradually reach steady state. In this experiment, we considered the wall temperature became stable when the variation of the thermocouple measurement in the cooper block was less than  $0.2^\circ\text{C}$  within one minute. The temperatures were calculated by averaging 30 measured values in the steady state. Bubble dynamics in the steady state were also recorded by the visualization system. The voltage was then increased to next predetermined value and the same recording procedure was repeated until the occurrence of a sudden increase of the measured temperature (temperature of the copper block increased over  $20^\circ\text{C}$  in one minute). Once the abrupt increase occurred, the power to the test heaters must be shut down immediately in order to protect the boiling surface from burnout. The critical heat flux (CHF) was determined as the heat flux just before the sudden increase of the measured temperature. It is a remarkable fact that all the experiments were conducted once the surfaces had been prepared. The aging effect of surfaces is not considered in present study. Additional studies on the roughness effects of pool boiling as a function of time will be performed in our future work.

#### 2.4. Calculations and uncertainty analysis

The heat flux through the surface can be calculated from the measured results of three pieces of axially aligned thermocouple using a one-dimension heat conduction model,

$$q = -k \frac{\Delta T}{\Delta x} \quad (1)$$

where  $k$  is the thermal conductivity of the copper,  $\Delta T$  is the temperature difference and  $\Delta x$  is the corresponding distance. Three different values of heat flux  $q$  can be calculated based on different combinations of the thermocouples. The variance of these values is negligible in the nucleate boiling regime. In order to lower the uncertainty of distance, the results based on  $T_1$  and  $T_3$  (as shown in Fig. 1(b)) are used in the present study. Therefore, Eq. (1) can be written as,

$$q = k \frac{T_1 - T_3}{\Delta x_{13}} \quad (2)$$

The wall temperature  $T_w$  can be then determined by  $T_3$  and heat flux  $q$ ,

$$T_w = T_3 - q \frac{\Delta x_{3w}}{k} \quad (3)$$

where  $\Delta x_{3w}$  is the distance between thermocouple  $T_3$  and the top wall of the copper block. The heat transfer coefficient  $h$  can be calculated as follows,

$$h = \frac{q}{\Delta T_{sat}} \quad (4)$$

in which  $\Delta T_{sat}$  is wall superheat and can be calculated by:

$$\Delta T_{sat} = T_w - T_{sat} \quad (5)$$

where  $T_{sat}$  is the liquid saturation temperature.

Uncertainties of the calculated parameters mainly come from temperature measurement error and position error of the thermocouples. The thermocouple with a diameter of 0.5 mm is inserted into an 0.6 mm diameter hole. Therefore, the maximum error for distance of two thermocouples is 0.1 mm. Assuming the conductivity has a relative error of 1%. Based on the methodology of Kline [25], the uncertainties of heat flux and surface temperature can be obtained by Eqs. (6) and (7),

$$\frac{\delta q}{q} = \sqrt{\left(\frac{\delta k}{k}\right)^2 + \left(\frac{\delta T_1}{T_1 - T_3}\right)^2 + \left(\frac{\delta T_3}{T_1 - T_3}\right)^2 + \left(\frac{\delta \Delta x_{13}}{\Delta x_{13}}\right)^2} \quad (6)$$

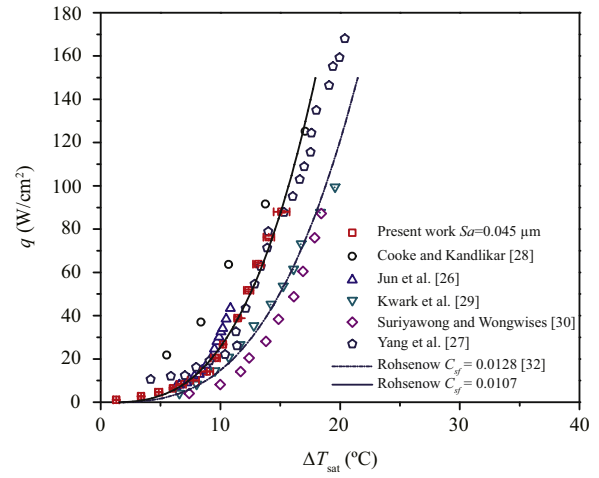


Fig. 6. Comparison of present experimental data with those in previous literature and Rohsenow correlation.

$$\delta T_w = \sqrt{\delta T_3^2 + \left(\frac{\Delta x_{3w}}{k} \delta q\right)^2 + \left(\frac{\Delta x_{3w} q}{k^2} \delta k\right)^2 + \left(\frac{q}{k} \delta \Delta x_{3w}\right)^2} \quad (7)$$

The estimated maximum uncertainty for heat flux and wall temperature were  $1.4 \text{ W/cm}^2$  and  $0.28^\circ\text{C}$ , respectively. The contact angle measurement had an accuracy of  $0.1^\circ$  according to the manufacturer. The surface roughness parameter was obtained by averaging five independent measurement data and provided deviation less than 20% refer to the nominal value.

### 3. Results and discussion

To validate the results of the present experiments, the experimental data of the reference surface ( $Sa = 0.045 \mu\text{m}$ ) are compared with previous studies which are performed in similar experimental conditions, i.e., saturated water pool boiling on smooth copper surface at atmosphere pressure. As shown in Fig. 6, the present experimental results are almost overlapped with the data from Jun et al. [26] and Yang et al. [27]. However, some discrepancy is observed in the study of other researchers [28–30]. This is because that nucleate boiling is such a complicated process which includes a lot of influence factors. The difference in any influence factor, measurement accuracy and data processing method can be a possible reason of the discrepancy. Generally, the present experimental data is quite comparable with the previous studies. Besides the previous data, the well-known Rohsenow correlation is also plotted in Fig. 6. The correlation was derived by Rohsenow in 1951 and was widely applied for predicting nucleate boiling heat transfer of saturated pool boiling [31].

$$\frac{C_p \Delta T_{sat}}{L} = C_{sf} \left[ \frac{q}{\mu L} \sqrt{\frac{\sigma}{g(\rho_l - \rho_v)}} \right]^{0.33} Pr^s \quad (8)$$

where  $C_p$  is the liquid specific heat,  $L$  is the latent heat of vaporization,  $\mu$  and  $Pr$  are the liquid dynamic viscosity and Prandtl number respectively. The parameter  $s$  is usually equal to 1 for water and 1.7 for the other fluids, and  $C_{sf}$  is a constant depending on the liquid-surface combination. For water-copper combination, the Rohsenow constant  $C_{sf}$  is usually evaluated to 0.0128, as suggested by Vachon et al. [32]. The data from Kwark et al. and Suriyawong and Wongwises can fit the Rohsenow correlation with  $C_{sf} = 0.0128$  well [29,30]. However, this correlation can not predict other data points in Fig. 6. A new Rohsenow correlation with  $C_{sf} = 0.0107$  is developed based on the present study and the data from Jun et al. and Yang et al. [26,27]. This new Rohsenow constant  $C_{sf} = 0.0107$

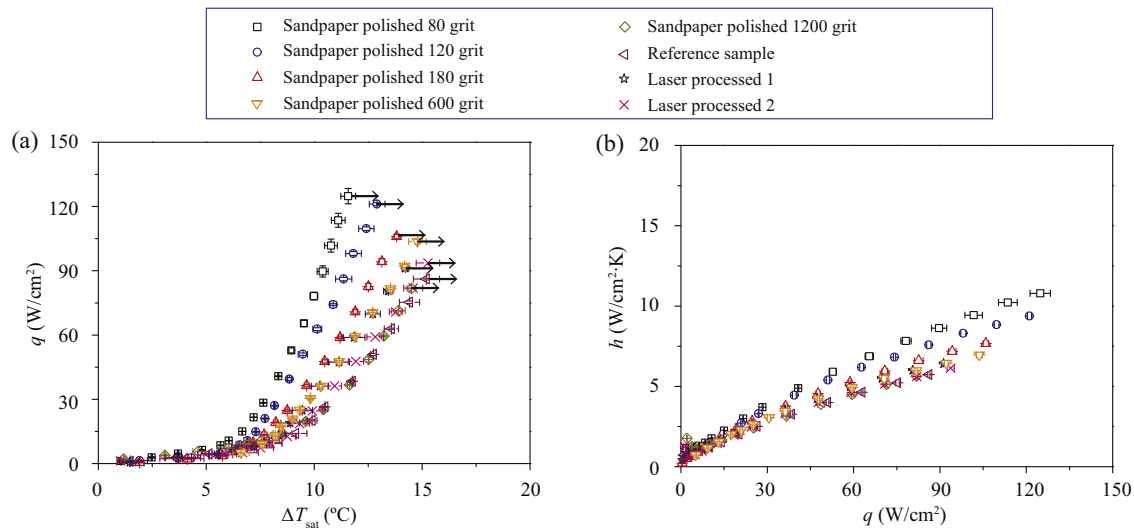


Fig. 7. Pool boiling curves of all the testing surfaces, (a) heat flux  $q$  as a function of wall superheat  $\Delta T_{sat}$ , (b) heat transfer coefficient  $h$  as a function of heat flux  $q$ .

can well predict the present data on reference surface and will be applied in the following discussion regarding correlation development.

Boiling curves for all the eight testing surfaces are shown in Fig. 7. It can be found that when the effective heat transfer area ratio  $r$  or  $Sa_p$  (corresponds to  $Pa$  in ISO 4287) is used to represent surface roughness (see also Table 1), there is not a clear trend of heat transfer coefficient with respect to surface roughness. Based on the experimental results, the same conclusion as that from the previous studies can be drawn, that is, the heat transfer coefficient is an increase function of surface roughness only under a consistent surface roughening technique. When two surfaces machined by different techniques are investigated in the pool boiling experiments, the one with smaller  $r$  or  $Sa_p$  is able to show better heat transfer performance (see sandpaper polished 180 grit and laser processed 1). However, if the waviness components are removed from the surface texture, as shown in Fig. 4. It can be easily observed that the boiling curves shift to the left as the surface roughness parameter  $Sa$  (corresponds to  $Ra$  in ISO 4287) increases, regardless of the surface preparation method. The trend can be seen more clearly in Fig. 7(b). Rougher surface has consistently higher heat transfer coefficient than the smoother one. Besides, the heat transfer coefficient increases as heat flux  $q$  becomes larger. A lot of researchers emphasized that surface roughness is meaningless without specifying the surface roughening technique, since they found that a smoother surface prepared by one surface roughening technique shows better heat transfer performance than a rougher one prepared by another surface roughening technique [14–17]. However, different conclusion can be drawn based on the present study. As we know, nucleation sites in pool boiling are in micro-scale. Some surface roughening techniques which increase the surface roughness parameter by enlarging large-scale surface texture will not provide more potential nucleation sites hence improving nucleation boiling heat transfer coefficient, just like the laser processed surfaces in the present study. Only when the same surface roughening technique is used, the one with larger  $Sa_p$  must provide more potential nucleation sites. The experimental data can effectively support this point of view. As shown in Fig. 8, three groups of data are picked from the boiling curves, i.e., sandpaper polished 80 grit, sandpaper polished 600 grit and femtosecond laser processed 1. The femtosecond laser processed 1 has similar  $Sa_p$  with the sandpaper polished 80 grit, while it has almost identical  $Sa$  with the sandpaper polished 600 grit after applying the standardized filtration process. It can be found that the femtosec-

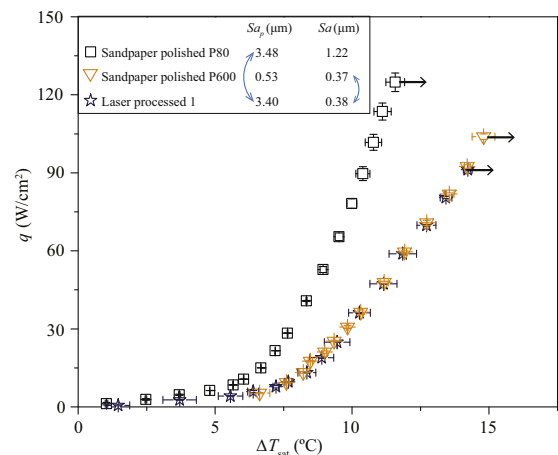


Fig. 8. Comparison of the experimental results of laser processed 1 with sandpaper polished 80 grit and sandpaper polished 600 grit.

ond laser processed 1 surface has overlapped boiling curve with the sandpaper polished 600 grit surface, while the sandpaper polished 80 grit surface shows much better pool boiling heat transfer performance. Obviously, the sandpaper polished 80 grit surface is “rougher” than the femtosecond laser processed 1 surface from the perspective of pool boiling, even though they have almost the same unfiltered roughness parameter  $Sa_p$ . It is inadequate to characterize the surface roughness by the unfiltered roughness parameters in pool boiling. The reason is that the unfiltered roughness parameters contain not only the characteristics of the micro-scale component (roughness), but also a part of the characteristics of the large-scale surface texture which is irrelevant to bubble nucleation. The standardized filtration process shown in Fig. 3 is able to eliminate the large-scale surface texture by applying the median denoising filter, the median smoothing filter, the areal gaussian filter with a cut-off of 25 μm and other operations suggested by ISO 3274 and ISO 25178. The filtered roughness parameter  $Sa$  can be a better way to characterize the surface roughness in pool boiling. It is reasonable to believe that the result in the present study is more close to the pure effect of the surface roughness by applying such filtered roughness parameter  $Sa$ . Besides, the choice of the cut-off value of the areal filter in the present study can be a possible benchmark for the unestablished standard of ISO 25178, at least

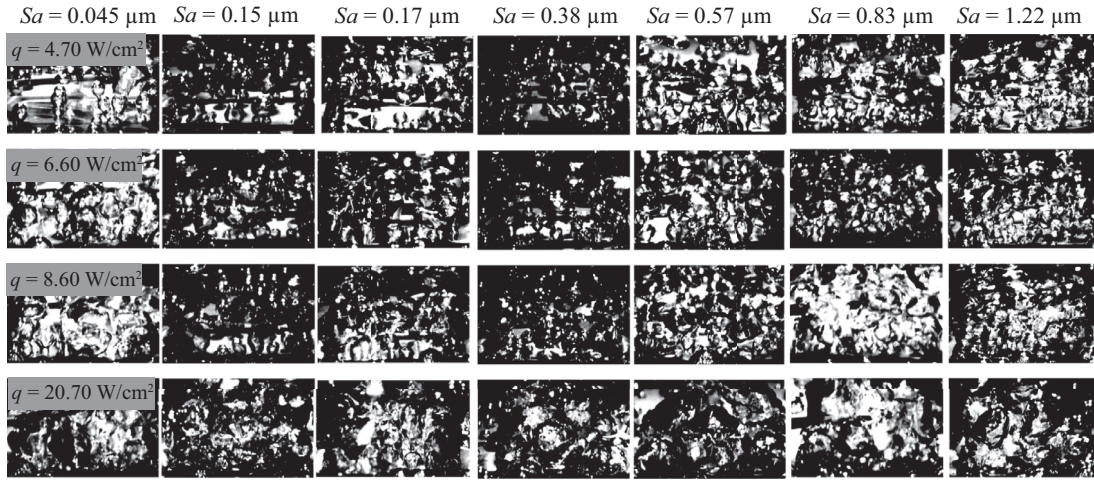


Fig. 9. Comparison of the visualized results with increasing  $Sa$ .

for surfaces applying in the pool boiling. Different with the surface roughness, the variation of surface wettability for hydrophobic surfaces in present study seems to have limited effect on the pool boiling heat transfer coefficient. The femtosecond laser processed 1 surface has overlapped boiling curve with the sandpaper polished 600 grit surface, although their contact angles are slightly different. Besides, the laser processed surface 2 ( $Sa = 0.15 \mu\text{m}$ ) and the sandpaper polished 1200 grits  $Sa = 0.17 \mu\text{m}$  also show the same result. On the other hand, the surface roughening technique is able to influence the CHF. In the unidirectionally polished surfaces, the CHF indeed becomes smaller as the surface roughness decreases. However, the smoothest unidirectionally polished one ( $Sa = 0.17 \mu\text{m}$ ) has a lower CHF than the reference surface (randomly polished) and the laser processed surface 2, even though it has a larger surface roughness parameter. Moreover, the CHF of two laser processed surfaces is almost identical (around  $93.5 \text{ W/cm}^2$ ), while their roughness parameters  $Sa_p$  and  $Sa$  are totally different. All the results demonstrate that the CHF is not a monotonic function of surface roughness. The surface wettability also plays an important role in determining the CHF.

The visualization studies of all the testing surfaces under a same heat flux are shown in Fig. 9 and the supplementary animation. For a certain testing surface, there is a larger number of nucleation bubbles as the heat flux increases. This explains why the nucleate boiling heat transfer coefficient  $h$  increases with the increasing heat flux  $q$ . It can be also found that the number of nucleation bubbles always increases with the roughness parameter  $Sa$  at a certain heat flux. However, it should be noted that the relation between roughness parameter  $Sa$  and number of nucleation sites is only a qualitative observation. Quantitative description of the potential nucleation sites can be received by size distribution of cavities on the surfaces, which has been conducted by some researchers [19,21]. The details will not be discussed here due to limited length. A simple quantitative analysis of bubble departure diameter and bubble departure frequency is conducted using the image processing program ImageJ. The results are shown in Fig. 10. Three surfaces with totally different surface roughening techniques and surface roughness parameters are included in the analysis. The heat flux is fixed at  $4.7 \text{ W/cm}^2$ . The results reveal that bubbles depart at a smaller diameter and higher frequency as the surface becomes rougher. This trend is consistent with that reported in Ref. [8]. The decreased bubble departure diameter and increased bubble departure frequency, together with the increased number of nucleation sites, result in the higher nucleate boiling heat transfer coefficient  $h$  of a rougher surface regardless of the surface roughening technique.

Eq. (8) developed by Rohsenow follows the relation  $q \propto \Delta T_{\text{sat}}^3$ . However, this relation is found invalid for some reported data. The exponent in the  $q - \Delta T$  relation should be determined based on the experimental conditions. Therefore, Eq. (8) is modified as follows,

$$\frac{C_p \Delta T_{\text{sat}}}{L} = C_{sf} \left[ \frac{q}{\mu L} \sqrt{\frac{\sigma}{g(\rho_l - \rho_v)}} \right]^{\frac{1}{n}} Pr^s \quad (9)$$

where  $n$  represents the exponent in the  $q - \Delta T$  relation. In order to perform the data analysis better, Eq. (9) is converted into the dimensionless form,

$$\frac{h \sqrt{\frac{\sigma}{g(\rho_l - \rho_v)}}}{k} = C_{sf}^{-n} \left( \frac{C_p \Delta T_{\text{sat}}}{L} \right)^{n-1} Pr^{1-sn} \quad (10)$$

In the visualized study of bubble dynamics, the bubble departure diameter is often found proportional to  $\sqrt{\frac{\sigma}{g(\rho_l - \rho_v)}}$  [33–36]. Therefore, the left hand side of Eq. (10) can be regarded as Nusselt number taking the bubble departure diameter as characteristic length. Eq. (10) can be then rewritten in the form of dimensionless numbers,

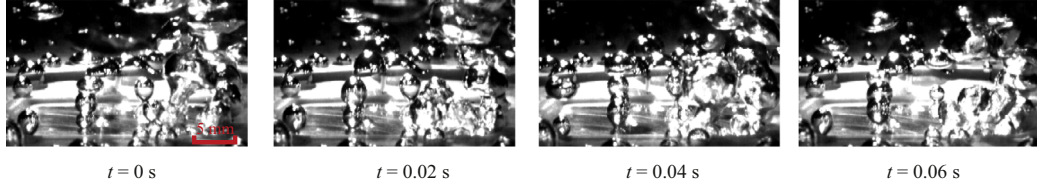
$$Nu = C Ja^{n-1} Pr^{1-sn} \quad (11)$$

where

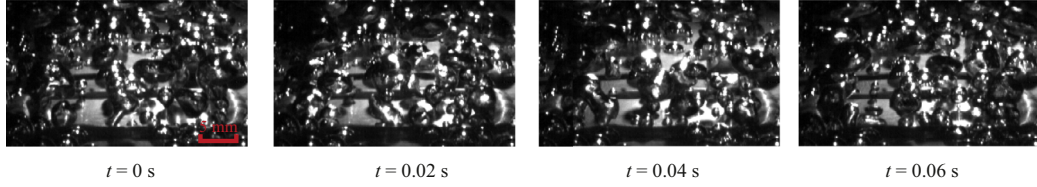
$$Ja = \frac{C_p \Delta T_{\text{sat}}}{L} \quad (12)$$

is the Jakob number, representing the ratio of sensible heat to latent heat absorbed during the boiling process.  $Nu$  is the Nusselt number of nucleation bubbles, and  $C$  is a constant depending on the Rohsenow constant  $C_{sf}$  and  $n$ . In Eq. (11),  $s$  is still 1 for water and 1.7 for the other fluids. But  $C$  and  $n$  are dependent on the experimental conditions. In the following analysis, they are correlated as a function of surface roughness  $Sa$  based on the experimental data. For convenience, the surface roughness  $Sa$  is also nondimensionalized to  $\frac{Sa}{Sa_0}$ , where  $Sa_0$  should be the representative surface roughness for typical metal surfaces.

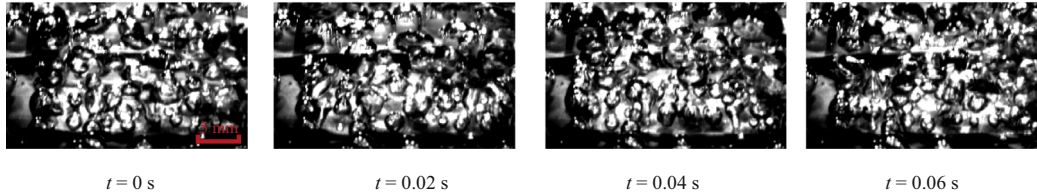
Following the suggestions of the other researchers in 2D profile ( $Ra$ ), the representatively areal arithmetical mean height  $Sa_0$  is also equal to  $0.4 \mu\text{m}$  [4,37]. The parameter  $n$  in the above equation is first evaluated by performing best fit from the experimental data. Fig. 11 shows the relation between  $Nu$  and  $Ja$  plotted on a logarithmic scale. The  $n - 1$  value of a given testing surface can be then determined as the slope of the straight line in the figure.  $n$  is then plotted as a function of  $\frac{Sa}{Sa_0}$  on the bottom right corner of

(a) Reference sample  $Sa = 0.045 \mu\text{m}$ 

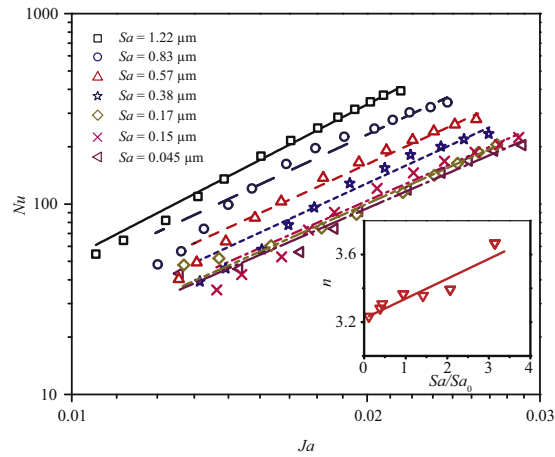
Average bubble departure diameter: 2.80 mm

(b) Femtosecond laser processed 1  $Sa = 0.38 \mu\text{m}$ 

Average bubble departure diameter: 2.27 mm

(c) Sandpaper polished 120 grit  $Sa = 0.83 \mu\text{m}$ 

Average bubble departure diameter: 1.82 mm

**Fig. 10.** A simple case in evaluating bubble departure diameter and bubble departure frequency. Please also see the supplementary animation.**Fig. 11.** Fitting and parameter development of the  $Nu - Ja$  curves.

**Fig. 11** and it is found to increase almost linearly with  $\frac{Sa}{Sa_0}$ ,

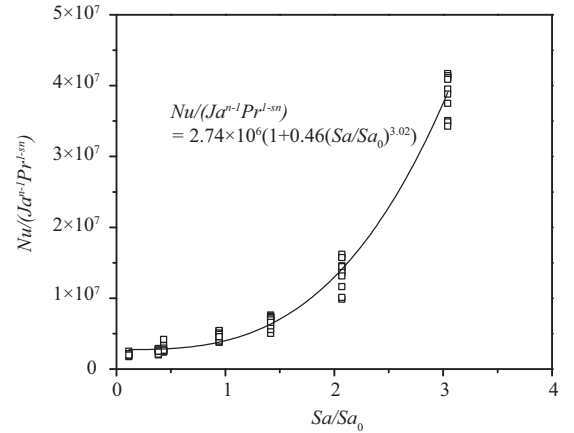
$$n = 3.22 + 0.12 \frac{Sa}{Sa_0} \quad (13)$$

and the correlation for  $C$  as a function of  $\frac{Sa}{Sa_0}$  can be then obtained by fitting the data points in **Fig. 12**.

$$C = 2.74 \times 10^6 \left(1 + 0.46 \left(\frac{Sa}{Sa_0}\right)^{3.02}\right) \quad (14)$$

We can summarize the final correlation in the dimensionless form,

$$Nu = C_1 \left(1 + C_2 \left(\frac{Sa}{Sa_0}\right)^{3.02}\right) Ja^{n-1} Pr^{1-sn} \quad (15)$$

**Fig. 12.** Evaluation of  $C$  by plotting  $Nu/Ja^{n-1} Pr^{1-sn}$  as a function of  $Sa/Sa_0$ .

where

$$C_1 = 2.74 \times 10^6 \quad (16)$$

$$C_2 = 0.46 \quad (17)$$

It should be noticed that **Eq. (15)** is likely to be valid for hydrophobic surfaces having surface roughness  $Sa$  in the range of  $0.045 \mu\text{m}$  to  $1.35 \mu\text{m}$ . Besides, the analysis of surface roughness  $Sa$  in present study is quite specific. It is not recommended to correlate other's experimental data with this equation, unless the same filtration process is followed in the surface roughness analysis. It should be also noticed that **Eq. (15)** can be only applied to predict the heat transfer coefficient in the nucleate boiling region. Before boiling occurs, the heat is mainly removed by natural convection,

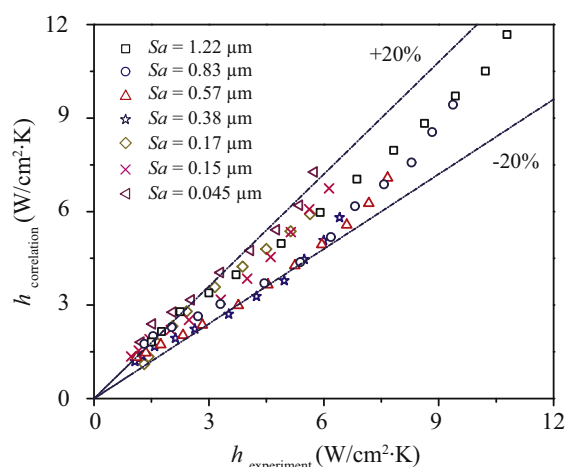


Fig. 13. Comparison between the experimental data and the predicted values by Eq. (15).

which is a totally different heat transfer mechanism with nucleate boiling. Using Eq. (15) to predict the heat transfer performance at this stage will cause large deviation. Fig. 13 is a comparison of the experimental data with the predicted value of Eq. (15) in the nucleate boiling region. The predicted values are in good agreement (within 20%) with the experimental data except for the low heat flux region (initial stage of bubble nucleation). In fact, the surface superheat at this region is very small, which results in a large relatively error of the surface superheat since the absolute error of temperature measurement is a fixed value. As a result, the uncertainty of the heat transfer coefficient at this region is also very large and the result can not be completely trusted. The inaccurate measurement is the most likely reason for the inconsistency between the experimental data and the predicted values at low heat flux region. Nevertheless, Eq. (15) is able to predict the heat transfer coefficient in the partially and fully developed nucleate boiling regime within an acceptable accuracy for different surface roughness  $Sa$ .

#### 4. Conclusions

The effect of surface roughness on the nucleate boiling heat transfer in a water pool was investigated. The copper surfaces used in the experiments were fabricated by three different surface roughening techniques (unidirectionally polished, randomly polished and laser processed) and had an average roughness in the range of 0.045  $\mu\text{m}$  to 1.35  $\mu\text{m}$ . The different surface roughening techniques could introduce different surface features and surface wettability. Bubble dynamics was also investigated to further understand the nucleate boiling heat transfer. Based on the results and discussion, the following conclusions were deduced:

- The unidirectionally polished surface had a lot of irregular scratches, while the femtosecond laser processed samples showed periodically fluctuating surface texture. For the sand-paper polished surfaces, the contact angle initially increased with surface roughness and remained a constant as the surface roughness further increased. The trend was a result of the spreading effect of the sessile droplet on the testing surfaces. Rougher surface with larger scratches could obstruct the droplet from spreading on the surface, especially in the direction normal to the scratches. Therefore, the static contact angle was increased as the surface became rougher. This increase did not occur anymore when the surface roughness  $Sa$  was larger than 0.57  $\mu\text{m}$ . Beyond which, the obstruction effect of the

droplet spreading had encountered a threshold value. A further increase in surface roughness would not cause the surface more hydrophobic. On the other hand, for the femtosecond laser processed surfaces, the static contact angles of both surfaces were around 153°. The surface roughness had negligible effect on the static contact angle.

- By applying a standardized filtration process on the measured surfaces, the boiling curves were found to shift to the left as the surface roughness parameter  $Sa$  became larger, regardless of the surface preparation method. The surface roughening technique had negligible effect on the nucleate boiling heat transfer coefficient once the large scale surface texture (surface form and waviness) were properly removed from the surface. On the other hand, the CHF was affected by both the surface roughness and surface roughening technique. It is not a monotonic function of surface roughness without specifying the surface roughening technique. The surface wettability also played an important role in determining the CHF.
- The rough surfaces enhanced the nucleate boiling heat transfer by providing more nucleation sites, decreasing bubble departure diameter and increasing bubble departure frequency. Besides, the number of nucleation bubbles was found to increase with heat flux, which well explained that the heat transfer coefficient is an increasing function of heat flux in all boiling curves.
- A correlation was developed based on the best fitting of the experimental data. The validation results showed that the correlation was able to well predict the heat transfer coefficient at different surface roughness  $Sa$ , especially in the fully developed nucleate boiling region.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### CRediT authorship contribution statement

**Simiao Fan:** Conceptualization, Software, Data curation, Formal analysis, Writing - original draft, Writing - review & editing, Validation. **Lishi Jiao:** Conceptualization, Software, Data curation, Writing - original draft, Validation. **Kai Wang:** Conceptualization, Software, Data curation, Writing - original draft, Validation. **Fei Duan:** Conceptualization, Software, Formal analysis, Writing - original draft, Writing - review & editing, Validation.

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#### Supplementary material

Supplementary material associated with this article can be found, in the online version, at [10.1016/j.ijheatmasstransfer.2020.120054](https://doi.org/10.1016/j.ijheatmasstransfer.2020.120054).

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